

STQ06214 Assignment

1. (a) From the data:

Smallest value = 131.4

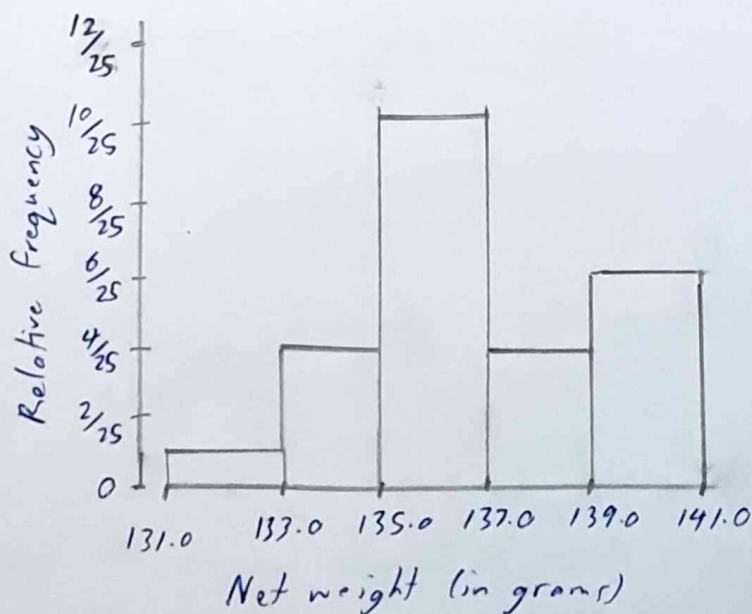
Largest value = 140.2

~~Suppose we use 5 classes with equal~~

approximated class width = $\frac{140.2 - 131.4}{5} = 1.76$

Round up to 2 for class width

Net weight (in grams)	Frequency	Relative frequency
$131.0 \leq x < 133.0$	1	$\frac{1}{25}$
$133.0 \leq x < 135.0$	4	$\frac{4}{25}$
$135.0 \leq x < 137.0$	10	$\frac{10}{25}$
$137.0 \leq x < 139.0$	4	$\frac{4}{25}$
$139.0 \leq x < 141.0$	6	$\frac{6}{25}$



~~$$P(X \leq 136) \rightarrow$$~~

$$\begin{aligned} (b) \quad P(X \leq 136) &= \frac{1}{25} \times 2 + \frac{4}{25} \times 2 + \frac{10}{25} \times 1 \\ &= \frac{20}{25} = \frac{4}{5} \end{aligned}$$

$$(c) \quad Y \sim N(136.6, 2.3^2)$$

$$\begin{aligned} P(Y \leq 136) &= P\left(Z \leq \frac{136 - 136.6}{2.3}\right) \\ &= P(Z \leq -0.26) \\ &= P(Z \geq 0.26) \\ &= 0.3974 \end{aligned}$$

Probability in (b) > (c). However, probability from (b) is from the sample data which is calculated from the histogram meanwhile, probability from (c) is assuming that the weights of the potato chips packet has normal distribution.

$$(d) \quad \bar{Y} \sim N(136.6, \frac{2.3^2}{25})$$

$$H_0: \mu = 136.6$$

$$H_1: \mu > 136.6$$

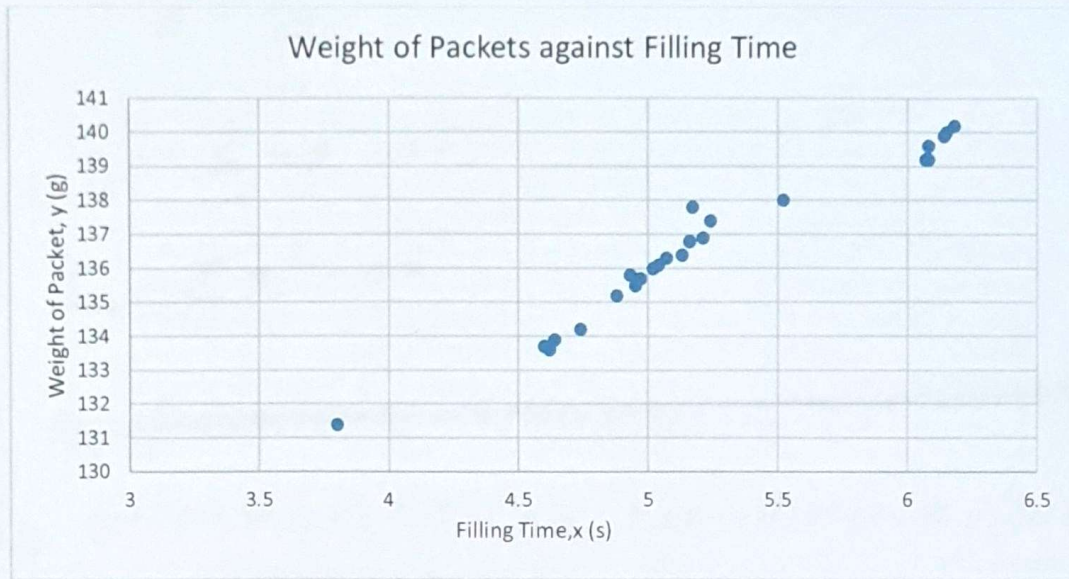
$$\bar{y} = \frac{\sum y}{n} = \frac{3416.8}{25} = 136.672$$

$$\begin{aligned} z &= \frac{\bar{y} - \mu_0}{\sigma/\sqrt{n}} = \frac{136.672 - 136.6}{2.3/\sqrt{25}} \\ &= \frac{0.072}{0.46} \\ &= 0.1565 \end{aligned}$$

$$Z_{0.05} = 1.645$$

Since $z < Z_{0.05} = 1.645$, we fail to reject H_0 . Hence, the claim that the average weight of potato chips packet is greater than 136.6g is false.

(c) i)



From the scatterplot above, we can see that it has a pattern which ~~is a~~ can be seen as a positive linear slope. Therefore, a simple linear model can be used for this scatterplot.

$$(ii) \quad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$$S_{xy} = \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}$$

$$S_{xx} = \sum_{i=1}^n x_i^2 - n \bar{x}^2$$

~~$$S_{xy} = [(4.74)(134.2) + (5.13)(136.4) + \dots + (6.14)(139.9) + (5.02)(136)]$$~~

$$S_{xy} = [(4.74)(134.2) + (5.13)(136.4) + \dots + (6.14)(139.9) + (5.02)(136)] - 25 \left(\frac{130.91}{25} \right) \left(\frac{3416.8}{25} \right)$$

$$= 17924.366 - 17891.73152$$

$$= 32.63448$$

$$S_{xx} = [4.74^2 + 5.13^2 + \dots + 6.14^2 + 5.02^2] - 25 \left(\frac{130.91}{25} \right)^2$$

$$= 694.2773 - 685.497124$$

$$= 8.780176$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{32.63448}{8.780176} = 3.7168 \text{ (4 decimal places)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 136.672 - (3.71683656)(5.2364) = 117.2092 \text{ (4 decimal places)}$$

$$\hat{y} = 117.209 + 3.717x$$

$\hat{\beta}_1 > 0$, it means there is positive linear relationship between x and $\hat{y}$. In other words, as x increases, $\hat{y}$ increases (the slope goes up)

(iii)

$$R^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}}$$

$$S_{xy} = 32.63448$$

$$S_{xx} = 8.780176$$

$$S_{yy} = \sum_{i=1}^n y_i^2 - n\bar{y}^2$$

$$= (134.2^2 + 136.4^2 + \dots + 139.9^2 + 136^2) - 25 \left(\frac{3416.8}{25} \right)^2$$

$$= 467107.68 - 466980.8896$$

$$= 126.7904$$

$$R^2 = \frac{(32.63448)^2}{(8.780176)(126.7904)}$$

$$= \frac{1065.009285}{1113.242027}$$

$$= 0.95667$$

R^2 value close to 1, which means the model is good and acceptable.
The fitted model explain 95.7% of the variation.

(iv)

$$\alpha = 0.05$$

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

$$se(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

$$\begin{aligned}\hat{\sigma}^2 &= \frac{SSE}{n-2} = \frac{S_{yy} - \hat{\beta}_1 S_{xy}}{n-2} \\ &= \frac{126.7904 - (3.716836656)(32.63448)}{25-2} \\ &= \frac{126.7904 - 121.2970315}{23} \\ &= 0.238842\end{aligned}$$

$$\begin{aligned}se(\hat{\beta}_1) &= \sqrt{\frac{0.238842}{8.780176}} \\ &= 0.164931549\end{aligned}$$

$$\begin{aligned}t &= \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} \\ &= \frac{3.716836656}{0.164931549} \\ &= 22.5356\end{aligned}$$

$$t_{0.025, 23} = 2.069$$

Since $t > t_{0.025, 23}$, we reject H_0 at $\alpha = 0.05$.

Therefore, there is linear relationship between filling time, x and weight of packet, y . In other word, filling time, x is significant.

$$(v) \quad \hat{y} = 117.209 + 3.717x$$

for $x = 5$,

$$\begin{aligned}\hat{y} &= 117.209 + 3.717(5) \\ &= 135.794\end{aligned}$$

for $x = 20$,

$$\begin{aligned}\hat{y} &= 117.209 + 3.717(20) \\ &= 191.549\end{aligned}$$

When $x = 5$, $\hat{y} = 135.794$. This predicted $\hat{y}$ value is acceptable since ~~the~~ x value is within the range ~~of the~~ $3.8 < x < 6.14$. However, when $x = 20$, the estimation $\hat{y}$ value is not reliable since x -value (20) is very far from the range $3.8 < x < 6.14$. Hence, using the fitted regression linear model is not appropriate when $x = 20$.